Q1 Perfect wala question

```
#!/bin/bash
echo "enter the perfect number"
read num
sum=0
for((i=1;i<num;i++))
do
  if((num%i==0))
then
    sum=$((sum+i))
    fi
done
if((sum==num))
then
  echo "This is perfect number"
else
  echo "Not Perfect"
fi
```

Q2 Factorial wala

```
#!/bin/bash
echo " Enter the number"
read n
f=1
for((i=1;i<=n;i++));
 do
  f=$((f*i));
 done
echo " The number of Factorial is $f"
```

Q3 Maximumof three number

```
#!/bin/bash
echo "enter the first number"
read a
echo "enter the second number"
read b
echo "enter the third number"
read c
if((a>=b && a<=c))
then
echo "Maximum number is $a"
elif((b>=a && b>=c))
then
echo "Maximum number is $b"
else
echo "Maximumnumber is $c"
fi
```

Q4 Countof words in a sentence

```
#!/bin/bash
echo "Enter the sentence"
read sentence
word_count=$(echo $sentence | wc -w)
echo "The words in a sentence is $word_count"
```

Q5 pattern

0

0 1

0 1 2

0 1 2 3

```
#!/bin/bash
for i in {0..3}
 do
   for j in $(seq 0 $i)
   do
     echo -n "$j"
   done
   echo
 done
```

Q6 array of fruits

```
#!/bin/bash
fruits=("Apple" "Banana" "Cickoo")
echo "All elemets in he array"
echo "${fruits[@]}"
echo "printing each elements usingloop:"
for fruit in "${fruits[@]}"
do
echo "$fruit"
done
```

Q7 count of lines.words,character

```
  GNU nano 8.7                    file.sh
#!/bin/bash
echo "Enter the file"
read file
if [ -f "$file" ]; then
echo "the number of lines: $(wc -l < "$file")"
echo "the number of words: $(wc -w < "$file")"
echo "the number of character: $(wc -m < "$file")"
```

```
else
  echo "Invalid"
fi
```